

Prior Sensitivity in Bayesian Structural Equation Modeling for Sparse Factor Loading Structures

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Xinya Liang¹ 

Abstract

Bayesian structural equation modeling (BSEM) is a flexible tool for the exploration and estimation of sparse factor loading structures; that is, most cross-loading entries are zero and only a few important cross-loadings are nonzero. The current investigation was focused on the BSEM with small-variance normal distribution priors (BSEM-N) for both variable selection and model estimation. The prior sensitivity in BSEM-N was explored in factor analysis models with sparse loading structures through a simulation study (Study 1) and an empirical example (Study 2). Study 1 examined the prior sensitivity in BSEM-N based on the model fit, population model recovery, true and false positive rates, and parameter estimation. Seven shrinkage priors on cross-loadings and five noninformative/vague priors on other model parameters were examined. Study 2 provided a real data example to illustrate the impact of various priors on model fit and parameter selection and estimation. Results indicated that when the 95% credible intervals of shrinkage priors barely covered the population cross-loading values, it resulted in the best balance between true and false positives. If the goal is to perform variable selection, a sparse cross-loading structure is required, preferably with a minimal number of nontrivial cross-loadings and relatively high primary loading values. To improve parameter estimates, a relatively large prior variance is preferred. When cross-loadings are relatively large, BSEM-N with zero-mean priors is not recommended for the estimation of cross-loadings and factor correlations.

Keywords

Bayesian structural equation modeling, prior sensitivity, sparse factor loading structure, variable selection, Bayesian model fit

¹University of Arkansas, Fayetteville, AR, USA

Corresponding Author:

Xinya Liang, Educational Statistics and Research Methods, Department of Rehabilitation, Human Resources and Communication Disorders, University of Arkansas, 751 West Maple Street, Fayetteville, AR 72701, USA.

Email: xl014@uark.edu

Bayesian structural equation modeling (BSEM) has become increasingly popular for analyzing complex factor structures due to its ability to incorporate exploratory features into traditional structural equation models (SEMs). In multidimensional measurement scales, cross-loadings (loadings on nontarget factors) are commonly encountered and routinely fixed to zero if they are smaller than a target threshold (e.g., threshold = .3; Cudeck & O'Dell, 1994). BSEM is flexible on the exploration and estimation of a *sparse* factor loading structure that produces sparse cross-loadings. That is, in a sparse factor loading matrix, most cross-loading entries are zero, and only a few important cross-loadings are nonzero (Trendafilov & Adachi, 2015; Zou et al., 2006).

BSEM can assign informative priors on small-sized parameters such as cross-loadings, and allow for free estimation of all these parameters. The informative priors are also known as the shrinkage priors that aim at shrinking trivial parameter estimates toward zero, with a goal to yield a parsimonious factor structure while retaining reasonable parameter estimation for nontrivial effects. The shrinkage priors can take various forms of distributions, including the normal distribution, Laplace distribution (Park & Casella, 2008; Tibshirani, 1996), scale mixture of normals (e.g., Carvalho et al., 2010), and more. The term *BSEM* refers to a broad range of Bayesian analyses in SEM with various prior distributions. In this study, the investigation was focused on the BSEM with small-variance normal distribution priors (BSEM-N) proposed by Muthén and Asparouhov (2012). BSEM-N has been applied in many empirical studies (e.g., De Bondt & Van Petegem, 2015; Price, 2017; Reis, 2019) due to its relatively convenient implementation using *Mplus* (Muthén & Muthén, 1998-2010).¹ Given the increasing prevalence of BSEM-N with multidimensional data, the prior sensitivity in BSEM-N as a variable selection and estimation method is worthy of investigation.

Bayesian Estimation

Bayesian estimation treats parameters as random variables with certain distributions that are assessed through the posterior distributions of parameters $p(\boldsymbol{\theta}|\mathbf{y})$ formed by updating the prior knowledge $p(\boldsymbol{\theta})$ given the data likelihood $p(\mathbf{y}|\boldsymbol{\theta})$, expressed as

$$p(\boldsymbol{\theta}|\mathbf{y}) \propto p(\boldsymbol{\theta})p(\mathbf{y}|\boldsymbol{\theta}), \quad (1)$$

where $\mathbf{y}$ is a vector of observed indicators and $\boldsymbol{\theta}$ is a vector of parameters. The choice of priors is critical in Bayesian estimation. Priors can range from noninformative (no or little prior knowledge) to informative (strong prior belief). Noninformative priors typically come with large prior variances and have a minor impact on the formation of posteriors. Informative priors represent a strong belief of the parameters and usually have small variances. Imposing meaningful informative priors may drag the posteriors away from the likelihood of improper data due to, for example, small sample sizes and/or nonnormal data (Finch & Miller, 2019; Lee, 2007; Liang, 2014; Liang & Yang, 2014).

Bayesian estimation is commonly accomplished by iterative sampling of a large number of observations from the posterior distributions of unknown parameters using the Markov chain Monte Carlo (MCMC) algorithm; one example is the Gibbs sampler (Geman & Geman, 1984). The convergence of multiple chains can be determined using the proportional scale reduction (PSR; e.g., Gelman & Rubin, 1992) procedure. A PSR factor quantifies the ratio of the between-chain variation to the within-chain variation. A PSR factor close to 1 indicates a good convergence of the estimation. On convergence, statistical properties (e.g., mean, median, quantiles) of the posterior distributions can be used as Bayesian inferences of parameters.

Prior Specification in BSEM-N

A general factor analysis model with continuous indicators can be written as

$$\mathbf{y} = \mathbf{v} + \mathbf{\Lambda}\boldsymbol{\xi} + \boldsymbol{\delta}, \quad (2)$$

where $\mathbf{v}$ is a $p \times 1$ vector of item intercepts, $\mathbf{\Lambda}$ is a $p \times q$ matrix of factor loadings, $\boldsymbol{\xi}$ is a $q \times 1$ vector of latent factors, and $\boldsymbol{\delta}$ is a $p \times 1$ vector of measurement errors typically assumed independent and normally distributed. Let λ_{jk} be the factor loading for item j measuring latent factor k . BSEM-N assigns a conditional Gaussian (normal) distribution $N(\mu_{jk}, \sigma_{jk}^2)$ on each element λ_{jk} in the factor loading matrix $\mathbf{\Lambda}$ as follows:

$$p(\mathbf{\Lambda} | \boldsymbol{\Sigma}_0, \mathbf{K}_0) \propto \exp \left(\sum_{j=1}^p \sum_{k=1}^q -\frac{1}{2\sigma_{jk}^2} (\lambda_{jk} - \mu_{jk})^2 \right), \quad (3)$$

where μ_{jk} and σ_{jk}^2 are the hyperparameters that represent the mean and variance of the factor loading, and $\mathbf{K}_0$ and $\boldsymbol{\Sigma}_0$ are matrices comprising all μ_{jk} and σ_{jk}^2 , respectively. The hyperparameters (μ_{jk} and σ_{jk}^2) can be given fixed values or hyper prior distributions. The former is used for the BSEM-N prior specifications in *Mplus*.

With a focus on the sparse factor loading matrix, to investigate the factor loading pattern, the factor loading matrix $\mathbf{\Lambda}$ can be partitioned into a group of primary factor loadings (specified *a priori*) and a group of cross-loadings (exploratory). BSEM-N explores the factor loading pattern by imposing different prior distributions on each group of loadings. Specifically, primary loadings are estimated in a confirmatory manner; each is assigned noninformative priors (e.g., $N(0, 10^{10})$) or accurate (or weakly) informative priors if prior knowledge is available from past experiences. Cross-loadings are estimated in an exploratory manner; they are assigned shrinkage priors with a mean of zero and a small variance (e.g., $N(0, .01)$). The aim of using shrinkage priors is to penalize trivial cross-loading estimates toward zero while leaving nontrivial effects freely estimated. This way, cross-loadings that are set to zero in traditional SEM are allowed a small variation near zero through the manipulation of priors. To perform the cross-loading selection, the significance of the cross-loadings can be assessed based on the 95% credible interval (CI) (95% CI not covering zero indicates significance of the estimate), the Wald test (Wald, 1945), or the likelihood

ratio test (Neyman & Pearson, 1933). The 95% CI in Bayesian estimation is an interval that covers parameter values between 2.5 and 97.5 percentiles of the posterior distribution.

The prior variance controls the shrinkage of parameter estimates, and should not be too large or too small. With a large variance, priors have little influence and the data likelihood dominates the formation of the posteriors. This leads to posterior estimates close to frequentist estimates; yet potential model underidentification² may occur in complex models. With a small prior variance, posterior distributions are pulled toward the prior distributions and allowed only limited variabilities. If the prior mean was set at zero (as what is typically used in shrinkage normal priors), this could worsen the parameter estimates (especially for nontrivial effects). With that said, the small prior variances serve as “soft” constraints that help perform variable selection and identify models that may not be estimable using traditional SEM.

Given a sparse loading structure, research has typically been focused on manipulating the shrinkage priors placed on cross-loadings. Nevertheless, the noninformative or vague priors assigned to other model parameters (intercepts, loadings, error variances) may also impact the modeling results. van Erp et al. (2018) found that default priors (noninformative or vague) from different software programs could lead to fairly different parameter estimates in a full SEM model, especially with small sample sizes. A sensitivity test that considers various noninformative/vague priors (in addition to the examination of choices of shrinkage priors) is of interest to assess the robustness of modeling results to the prior choices. Although parameter estimates could be sensitive to default prior choices, it is currently unknown what the influence of noninformative/vague priors is on variable selection using BSEM-N with shrinkage priors, which is worth an exploration.

Model Fit Evaluation

A popular procedure for evaluating Bayesian model fit is using the posterior predictive checking (PPMC), where a posterior predictive p value (PP p ; Meng, 1994) is computed as the proportion of iterations where the fit function calculated based on the observed data is smaller than that based on the replicated data. An arbitrary fit function can be used such as the chi-square fit function. A PP p value around .5 indicates a good model fit. A PP p value smaller than .05 or .01 indicates a potential model misspecification (Muthén & Asparouhov, 2012). The PPMC has been shown to better control the false positives and to be less sensitive to trivial model misfits when compared with chi-square tests in frequentist methods (e.g., Lee, 2007). However, with small-variance priors, the null hypothesis specifies small and irrelevant differences between the parameter value and zero, comparing with the traditional null hypothesis that states no difference between the parameter value and zero. In this case, the PP p has been shown to provide inconsistent evidence and be less powerful than traditional tests (e.g., chi-square tests) unless a large sample size is reached. As an alternative, the prior–posterior predictive p value (PPP p ; Hoijsink &

van de Schoot, 2018) has been proposed to test small-variance priors, which uses the prior distributions of parameters directly on computing the $PPPp$. In contrast, the effect of priors on PPp is mediated through the posterior distribution. The $PPPp$ has been shown to be more powerful than PPp and perform well in the context of simple regression model. Nonetheless, the generalization of $PPPp$ to structural equation models still warrants an investigation.

Moreover, model goodness-of-fit indices and information criteria (IC) can also be used for model fit evaluation. Recent advancements have proposed several Bayesian variates of fit indices, including Bayesian root mean square error of approximation (BRMSEA; Hoofs et al., 2018), Bayesian comparative fit index (BCFI), and Bayesian Tucker–Lewis index (BTLI; Garnier-Villarre & Jorgensen, 2019). These Bayesian fit indices are computed based on the difference between observed and replicated discrepancy functions at each MCMC iteration, similarly to that in the PPMC procedure. As information from each sample draw in the entire posterior distribution is used in the computation, one benefit is that the uncertainty of Bayesian fit indices can be assessed. BRMSEA has been shown advantageous in large-sample Bayesian confirmatory factor analysis (CFA; Jöreskog, 1969). Conventional guidelines for RMSEA less than .08 (Browne & Cudeck, 1992) may also apply to BRMSEA with a sample size greater than 1,000 (Hoofs et al., 2018). The use of fixed cutoff values for BCFI and BTLI, however, is not recommended in Garnier-Villarre and Jorgensen (2019), because the cutoffs are not generalizable to all sample characteristics. Similar issues with using fixed cutoffs are also discussed in frequentist SEM (e.g., Fan & Sivo, 2005; Kite et al., 2018).

On the other hand, the comparative fit between two models can be evaluated using information criteria such as Bayesian information criterion (BIC; Schwarz, 1978) and deviance information criterion (DIC; Spiegelhalter et al., 2002). A smaller IC value indicates better model fit. Asparouhov et al. (2015) recommended using DIC over BIC for comparing among BSEM-N models or between BSEM-N and CFA models. Hoijtink and van de Schoot (2018) found that the PPp and DIC were inconsistent with small-variance priors.

Practical Considerations

It is worth noting that there is an estimator bias-variance trade-off (Friedman, 1997) when zero-mean shrinkage priors are used. In sparse models where only a few parameters are important, one desire of using shrinkage priors is to decrease the variances and reduce the model complexity. However, as the shrinkage priors are applied universally to all trivial and nontrivial effects, the increase in shrinkage (i.e., the use of smaller prior variances) not only penalizes trivial effects more heavily toward zero, but concurrently inducing more bias in nontrivial parameter estimates (see, Jacobucci et al., 2019). An alternative would be to use shrinkage priors to guide variable selection to identify nontrivial cross-loadings for further model modifications.

If the model contains many nonzero cross-loadings, indicating a *dense* factor structure, the selection of nontrivial cross-loadings may be of less interest, and a more important goal would be to improve the parameter estimation. In this case, weakly informative or informative priors with a nonzero prior mean (reflecting the distribution of cross-loadings) may be considered to improve the model estimation. Having said that, one potential risk of using weakly informative priors is that the model can become nonidentified at a certain point when many cross-loadings are freely estimated. Moreover, if many nontrivial cross-loadings are present, it may be indicative of the need for scale revisions and revalidations. Therefore, the choice of priors depends on the primary research interest (variable selection, model estimation, or both).

Prior Research and the Goal of Current Study

Several simulation studies applying BSEM-N to analyze models containing cross-loadings have focused on the evaluation of parameter estimation, including examining the estimation bias, 95% coverage rates and power. Guo et al. (2019) examined the performance of BSEM-N and exploratory SEM (ESEM; Asparouhov & Muthén, 2009) with dense cross-loading structures. All indicators had nonzero cross-loadings, where the magnitudes of nontrivial cross-loadings ranged from .1 to .4 and trivial cross-loadings were $\pm .01$. The priors examined included zero-mean priors of $N(0, .005)$, $N(0, .01)$, and $N(0, .02)$, and a nonzero-mean prior of $N(.1, .01)$ that reflected the estimate of the cross-loading mean. The examination considered only sample sizes of 200 and more and high primary loadings of .8. Xiao et al. (2019) compared BSEM-N with ESEM using dense factor structures with only one sample size of 500. A correct prior (following the distribution of cross-loadings) and two zero-mean priors, $N(0, .01)$ and $N(0, .09)$, were examined. Both studies above did not consider sparse factor structures. BSEM-N was found to outperform independent cluster model (ICM; McDonald, 1985) of CFA in which all cross-loadings are fixed to zero. When the prior distribution was consistent with the distribution of cross-loadings, BSEM-N outperformed ESEM. Moreover, Liang et al. (2020) compared BSEM-N and ESEM in full SEM models with sparse factor structures when data were ordinal. The model had five latent factors with two exploratory factors and three confirmatory factors. Four zero-mean priors with variances from .005 to .03 were assigned to cross-loadings on the two exploratory factors, respectively. BSEM-N was outperformed by ESEM in large samples, but may be considered in small samples to overcome the convergence problem in complex models.

BSEM-N as a variable selection method (e.g., detect cross-loadings), however, has received less attention. Muthén and Asparouhov (2012) examined three-factor models with sparse cross-loading structures. This study considered sample sizes not exceeding 500 and high primary factor loadings of .8. In addition to investigating parameter estimates, the power of detecting cross-loadings was also reported but results were limited to only one shrinkage prior of $N(0, .01)$. In part of a study by Lu et al. (2016),

Table 1. Summary of Simulation Studies Using BSEM in Factor Analysis With Cross-Loadings.

Article	Factor structure	Sample size	Primary factor loading value	Cross-loading value	BSEM implementation
Muthén and Asparouhov (2012)	Three-factor model with three cross-loadings (sparse)	100, 200, 500	0.8	0.1, 0.2, 0.3	BSEM-N with noninformative priors (Mplus default) BSEM-N with priors on cross-loadings: $N(0, .01)$ BSEM-N with priors on cross-loadings: $N(0, .001)$, $N(0, .01)$ and $N(0, .1)$ BSEM with spike-and-slab priors (SSPs) on cross-loadings
Lu et al. (2016)	Three-factor model with one or three cross-loadings (sparse)	100, 200, 300	1	0.1, 0.2, 0.3	BSEM-N with priors on cross-loadings: $N(0, .001)$, $N(0, .01)$ and $N(0, .1)$ BSEM with spike-and-slab priors (SSPs) on cross-loadings
Guo et al. (2019)	Three-factor model with dense loading matrix	200, 500, 1,000	0.8	Size of nontrivial cross-loadings: 0.1 to 0.4 Size of trivial cross-loadings: 0.01	BSEM-N with priors on cross-loadings: $N(0, .005)$, $N(0, .01)$, $N(0, .02)$ and $N(1, .01)$
Xiao et al. (2019)	Two- or five-factor models with dense loading matrix	500	Primary loadings follow a normal distribution with a mean of 0.6, and a variance of 0.01, 0.04, or 0.09	Cross-loadings follow a normal distribution with a mean of 0.1, or 0.2, and a variance of 0.01, 0.04, 0.09	BSEM-N with priors on cross-loadings: $N(0, .01)$ and $N(0, .09)$

Note. BSEM = Bayesian structural equation modeling. Only simulation settings using BSEM are summarized in the table. Select studies compared BSEM with other methods such as exploratory SEM.

they compared BSEM-N with a modern prior, the spike and slab prior (SSP; Mitchell & Beauchamp, 1988), using each as variable selection methods for exploring sparse factor loading structures. Normal priors with a variance of .001, .01, or .1 were applied in BSEM-N. Results of true and false positive rates for cross-loading detection showed that BSEM-SSP was superior to BSEM-N only with relatively large sample sizes (e.g., 300) and cross-loadings (e.g., .3). BSEM-N with a proper prior variance may perform comparably to or better than BSEM-SSP with relatively small samples and cross-loadings. Lu's study highlighted the important influences of prior variances on variable selection results in BSEM-N. A summary of past simulation study characteristics using BSEM in factor analysis is provided in Table 1.

Based on the discussion above, previous studies that used BSEM-N as a variable selection method for detecting nontrivial cross-loadings are limited. Most research assumed a highly reliable instrument, and none of that research has investigated the model using weak primary loadings generated from a less reliable instrument. The impact of various sample sizes and low to moderate primary loadings on model estimation results is not clear. The sensitivity to various prior choices has not been systematically investigated in terms of both parameter selection and estimation. While shrinkage priors were the primary interest in previous studies, no research has examined the sensitivity of results to noninformative/vague priors assigned to other model parameters using BSEM-N. In addition, as the recent development of some Bayesian fit measures (e.g., PPP p , BCFI, BTLI, BRMSEA), the sensitivity of Bayesian fit measures to various prior choices and factor structures warrants an investigation.

Given the rising popularity of BSEM-N, the current study aims at providing insights on the performance of BSEM-N as a tool for both variable selection and estimation in factor analysis models with sparse loading structures. A simulation study (Study 1) and an empirical study (Study 2) were conducted to explore the prior sensitivity in BSEM-N. Specifically, Study 1 examined the prior sensitivity in BSEM-N based on the model fit, population model recovery, variable selection performance, and parameter estimation. In addition to examining the sensitivity to shrinkage priors, the sensitivity to noninformative/vague priors was also investigated. Study 2 provided a real data example to illustrate the impact of various priors on model fit and parameter selection and estimation.

Study 1: Simulation

Data Generation Conditions

Data generation was based on a CFA model with two latent factors (ξ_1, ξ_2). Each latent factor was measured by five continuous indicators. Three factor loading matrices with one, two, and four cross-loadings were examined, respectively, as follows:

$$\text{Model 1} \quad \Lambda'_{\xi} = \begin{pmatrix} \lambda_p & \lambda_p & \lambda_p & \lambda_p & \lambda_p & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & c & \lambda_p & \lambda_p & \lambda_p & \lambda_p & \lambda_p \end{pmatrix}, \quad (4)$$

$$\text{Model 2} \quad \Lambda'_{\xi} = \begin{pmatrix} \lambda_p & \lambda_p & \lambda_p & \lambda_p & \lambda_p & c & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & c & \lambda_p & \lambda_p & \lambda_p & \lambda_p & \lambda_p \end{pmatrix}, \quad (5)$$

$$\text{Model 3} \quad \Lambda'_{\xi} = \begin{pmatrix} \lambda_p & \lambda_p & \lambda_p & \lambda_p & \lambda_p & c & c & 0 & 0 & 0 \\ 0 & 0 & 0 & c & c & \lambda_p & \lambda_p & \lambda_p & \lambda_p & \lambda_p \end{pmatrix}, \quad (6)$$

where λ_p denotes the primary loading and c denotes the cross-loading. For each loading matrix, primary loadings for all indicators were set to either .7 or .4, representing a relatively strong or weak association between the indicator and the latent factor (Garrido et al., 2016; Shi et al., 2019). Cross-loadings were set to either .1 or .3, considered to be of little importance or importance, respectively (Cudeck & O'Dell, 1994). Factor variances were fixed to 1 and the factor correlation r was fixed to .4. Error variances were set to $1 - \lambda_p^2 - c^2 - 2r\lambda_p c$, so all indicators were standardized to follow a normal distribution $N(0, 1)$.

Sample sizes were considered at seven levels: 50, 100, 200, 400, 600, 800, and 1,000. Small sample sizes were of interest due to the challenge of collecting sufficient data in education and psychology (Smid et al., 2019). Large sample sizes were considered which provided an opportunity to observe the change in outcomes as sample sizes increased. In Muthén and Asparouhov (2012), for a three-factor CFA model, even when sample sizes increased to 500 the power of detecting a low-magnitude cross-loading (i.e., .1) was less than .2. A wide range of sample sizes allowed for a systematic investigation of how sample sizes interact with the prior choices.

Accordingly, the prior sensitivity in BSEM-N was investigated under 84 data generation conditions: 7 sample sizes $\times$ 3 data generation models $\times$ 2 primary loadings $\times$ 2 cross-loadings.

Sensitivity Analysis for Shrinkage Priors

Each data set was analyzed using BSEM-N with seven shrinkage prior choices on cross-loadings. Though with the sparse cross-loading structures, all potential cross-loadings (i.e., ξ_1 by $y_6 - y_{10}$ and ξ_2 by $y_1 - y_5$) were assigned zero-mean shrinkage priors. The seven priors considered were $N(0, .001)$, $N(0, .005)$, $N(0, .01)$, $N(0, .02)$, $N(0, .03)$, $N(0, .05)$, and $N(0, .08)$, resulting in the 95% credible intervals equal to $\pm .06$, $\pm .14$, $\pm .20$, $\pm .28$, $\pm .34$, $\pm .44$, and $\pm .55$, respectively. When the prior variance was .001, the 95% CI ($\pm .06$) did not cover the cross-loadings of .1 and .3. This however would provide results similar to those from ICM-CFA, and has been recommended by Asparouhov et al. (2015) as a starting point in prior sensitivity analysis. Moreover, at least five priors are recommended to test for sensitivity with the gradual increase in the prior variance. For prior variances of .005 to .02, the 95 CIs ($\pm .14$ to $\pm .28$) covered the cross-loading of .1 but not .3. For prior variances of .03 to .08, the 95% CIs ($\pm .34$ to $\pm .55$) covered both cross-loadings of .1 and .3. This wide range of prior selections allowed for the examination of prior sensitivity in BSEM-N as a tool for both variable selection and estimation. Other model parameters were assigned noninformative priors using the *Mplus* default settings—a normal prior $N(0, \infty)$ for

primary loadings, an Inverse Wishart prior $IW(0, -3)$ for factor variances and covariances, and an Inverse Gamma (IG) prior $IG(-1, 0)$ for error variances. The data analysis conditions for the investigation of shrinkage prior sensitivity were 588 (84 data generation conditions $\times$ 7 shrinkage priors).

Sensitivity Analysis for Noninformative/Vague Priors

Because noninformative/vague priors have been shown to influence the model estimation especially with small sample sizes (van Erp et al., 2018), the sensitivity of parameter selection and estimation results to various noninformative/vague priors was also examined. Specifically, the noninformative/vague prior sensitivity was examined for Model 1 with the smallest sample size of 50, primary loading of .7, and one cross-loading of .3. The noninformative/vague priors assigned to item intercepts, factor loadings, and error variances were manipulated. For intercepts and loadings, noninformative normal priors were assigned, and for error variances, inverse Gamma priors were assigned. According to van Erp et al. (2018), the improper prior $IG(0, 0)$ and the vague priors such as $IG(.001, .001)$ that approximate this improper prior are not recommended due to major convergence problems. The examination thus included the following five noninformative/vague priors based on van Erp et al. (2018):

- Prior 1: $IG(-1, 0)$ and $N(0, \infty)$ (noninformative improper, *Mplus* default)
- Prior 2: $IG(-.5, 0)$ and $N(0, \infty)$ (noninformative improper)
- Prior 3: $IG(.01, .01)$ and $N(0, \infty)$ (vague proper)
- Prior 4: $IG(.1, .1)$ and $N(0, \infty)$ (vague proper)
- Prior 5: $IG(-1, 0)$, $N(0, 1000)$ for intercepts, and $N(0, 100)$ for loadings (vague normal)

For each of the seven shrinkage priors imposed on cross-loadings (i.e., prior variance = .001, .005, .01, .02, .03, .05, or .08), five noninformative/vague priors on other parameters (intercepts, loadings, error variances) were examined. The data analysis conditions for the investigation of noninformative/vague prior sensitivity were 35 (1 data generation condition $\times$ 7 shrinkage priors $\times$ 5 noninformative/vague priors).

BSEM-N Implementation

Normally distributed data were generated with 500 replications using SAS (SAS Institute Inc, 2002-2012) and *Mplus* 8.4 (Muthén & Muthén, 1998-2010). Data were fitted to the model considering all potential cross-loadings as model parameters. The variance of each factor was fixed to 1 for identification. Gibbs sampling was used for drawing posterior samples from two MCMC chains. Each MCMC chain was specified a minimum of 1,000 iterations³ and a maximum of 100,000 iterations. The

PSR factors less than 1.05 for all model parameters indicate the convergence. Upon convergence, the last half of the samples were used to summarize the posterior distribution. The median of the posterior distribution was used as the Bayesian point estimate. Sample *Mplus* code for implementing BSEM-N is provided in Appendix B.

Outcome Evaluation

Inadmissible solutions (nonconvergence, out-of-bound estimates) were examined and removed from further analysis. Model fit was evaluated using the *PPp* rejection rates and *PPPp* rejection rates. Both rejection rates were defined as the proportion of replications that rejected the analysis model at the .05 level. Moreover, Bayesian fit indices (BRMSEA, BCFI and BTLI) and Bayesian information criteria (BIC and DIC) were reported.

For variable selection, the model recovery rate, true positive rate, and false positive rate were evaluated. The model recovery rate was computed as the proportion of replications that correctly identified all nonzero cross-loadings in the data generation model. For example, if the data generation model had four cross-loadings, the recovery was considered successful if the analysis model correctly detected all four cross-loadings as significant. If more than or less than four cross-loadings were detected, the recovery was considered failed. The true positive rate was the proportion of nonzero cross-loadings that were concluded to be nonzero. The false positive rate was the proportion of zero cross-loadings that were concluded to be nonzero. The statistical significance of cross-loadings was determined based on the 95% credible interval, which, if not covering zero, indicates the significance of cross-loadings from zero. To obtain the 95% credible interval, all posterior samples from each replication were saved to an external file using *Mplus*. The 2.5 and 97.5 percentiles of the posterior distribution using the second half of the samples from the two chains were obtained as the lower and upper bounds of the 95% credible interval. For model parameter estimation, average posterior point estimates, 95% credible intervals, and 95% parameter coverage rates were reported for primary loadings, nonzero cross-loadings, and the factor correlation. The 95% coverage rate was defined as the proportion of replications in which the 95% credible interval covered the population parameter value.

Results for Shrinkage Prior Sensitivity

To conserve space, results of shrinkage prior sensitivity are presented for Model 1 (one cross-loading) and Model 3 (four cross-loadings). Results for Model 2 (two cross-loadings) are available in the online Supplemental Material.

Inadmissible Solutions and Model Fit. Inadmissible solution rates were zero given a minimum of 1,000 iterations with PSR smaller than 1.05 for all conditions. For model fit measures, an analysis of variance (ANOVA) with first-order interactions was conducted to identify salient factors that contributed the most to the variation in each fit

Table 2. Model Fit Measures.

Primary loading	Cross-loading	Prior variance	PPp rejection rate (%)	PPPp rejection rate (%)	BRMSEA	BCFI	BTLI	BIC	DIC
.4	.1	.001	1.42	1.57	.02	.94	.92	12,746	12,577
		.005	.58	.40	.02	.95	.93	12,743	12,577
		.01	.41	.16	.02	.95	.94	12,741	12,578
		.02	.32	.06	.02	.96	.94	12,740	12,579
		.03	.35	.01	.02	.96	.94	12,740	12,580
		.05	.30	.00	.02	.96	.94	12,739	12,581
		.08	.21	.00	.02	.96	.93	12,739	12,582
	.3	.001	19.02	20.16	.03	.93	.91	12,634	12,464
		.005	5.18	5.41	.03	.95	.93	12,627	12,459
		.01	1.90	.96	.02	.95	.94	12,623	12,457
		.02	.72	.25	.02	.96	.94	12,620	12,457
		.03	.49	.06	.02	.96	.95	12,619	12,457
		.05	.26	.02	.02	.97	.95	12,617	12,457
		.08	.29	.00	.02	.97	.95	12,617	12,459
	.7	.001	1.63	5.52	.02	.99	.99	11,365	11,198
		.005	.59	.37	.02	.99	.99	11,360	11,198
		.01	.55	.14	.02	.99	.99	11,359	11,199
		.02	.48	.03	.02	.99	.99	11,358	11,200
		.03	.54	.00	.02	.99	.99	11,358	11,200
		.05	.65	.00	.02	.99	.99	11,358	11,200
		.08	.54	.00	.02	.99	.98	11,358	11,200
	.3	.001	53.50	75.38	.06	.97	.96	10,953	10,786
		.005	3.65	62.54	.03	.99	.98	10,924	10,763
		.01	1.25	40.93	.02	.99	.99	10,920	10,760
		.02	.62	3.08	.02	.99	.99	10,918	10,759
		.03	.59	.22	.02	.99	.99	10,917	10,759
		.05	.50	.01	.02	.99	.99	10,917	10,759
		.08	.51	.00	.02	.99	.99	10,917	10,759

Note. PPp = posterior predictive p value; PPPp = prior-posterior predictive p value; BRMSEA = Bayesian root mean square error of approximation; BCFI = Bayesian comparative fit index; BTLI = Bayesian Tucker-Lewis index; BIC = Bayesian information criterion; DIC = deviance information criterion.

measure. The ANOVA contained four between-subject factors: model type, sample size, primary loading, and cross-loading, and one within-subject factor: prior variance. The important factors were selected based on the eta-square measure greater than .0588, indicating a moderate effect size (Cohen, 1973). The selected factors all showed statistical significance at the .05 level. As a result, the prior variance, primary loading and cross-loadings were identified as important factors for the model fit measures.

The model fit measures were summarized in Table 2 based on salient factors. Both the PPp and PPPp rejection rates were reasonably around or lower than the .05 level in most conditions. Except for conditions with cross-loadings of .3, when using a

prior variance equal to or smaller than .005, the rejection rate increased with a smaller prior variance. The *PPp* rejection rates with the prior variance of .001 could go up to an average of 53.50% when the primary loading was .7 across model types and sample sizes. In contrast, the *PPPp* rejection rates went up to 75.38%. Moreover, when the primary loading was .7 and prior variance was .01 or less, the *PPPp* rejection rates (40.93%-75.38%) were notably higher than the *PPp* reject rates (1.25%-53.50%), echoing Hoijtink and van de Schoot (2018), which showed that the *PPPp* was more powerful than the *PPp* with small-variance priors. Both evidences indicated that when nontrivial cross-loadings (.3) were constrained to be closer to zero due to the use of small-variance priors, the resultant model misfit may lead to frequent rejections of the model.

All Bayesian fit indices (BRMSEA, BCFI, BTLI) indicated well-fitting models based on the conventional cutoffs (RMSEA < .05 or .08, CFI < .95 or .90, TLI < .95 or .90; Browne & Cudeck, 1992; Hu & Bentler, 1998). However, it is to note that these cutoffs should be applied with caution, because the sampling distributions of fit indices vary depending on different sample characteristics and the cutoffs may not apply to all situations (Garnier-Villarreal & Jorgensen, 2019). When comparing across study conditions, larger primary loadings and larger prior variances yielded better model fit based on Bayesian fit indices. The Bayesian information criteria (BIC, DIC) also provided similar suggestions. A model was favored with a smaller BIC or DIC, associated with larger primary loadings and larger cross-loadings in general. However, one should be cautious with relatively small primary loadings (e.g., .4) and cross-loadings (e.g., .1), DIC may provide inconsistent suggestions from other fit measures.

Model Recovery, True Positives, and False Positives. Model recovery rates, true positive rates, and false positive rates are shown in Figures 1 to 3, respectively. In general, higher model recovery rates and higher true positive rates were associated with larger sample sizes, fewer numbers of nontrivial cross-loadings, and greater primary and cross-loading values. Prior variances between .005 and .05 appeared to provide reasonable model recovery and true positive rates under the current simulation conditions. Prior variances that were too small (<.005) or too large (>.05) did not result in good model recovery and true positive rates. Inflated false positive rates were observed in conditions when primary loadings were .7 and cross-loadings were .3. Especially in Model 3 with four cross-loadings, the false positive rates could go up to .92 with small prior variance of .001. False positive rates increased as the prior variance decreased.

When cross-loadings were .1, the model recovery and true positive rates were mostly low (<.1), except for Model 1 with primary loading of .7. In this condition, however, the recovery rates were no more than .48 and the true positive rates were no more than .51. The highest model recovery rates (around .90) were observed in Model 1 with a primary loading of .7 and cross-loading of .3. When the prior variance was .01 or smaller, as sample sizes increased, the recovery rates showed an

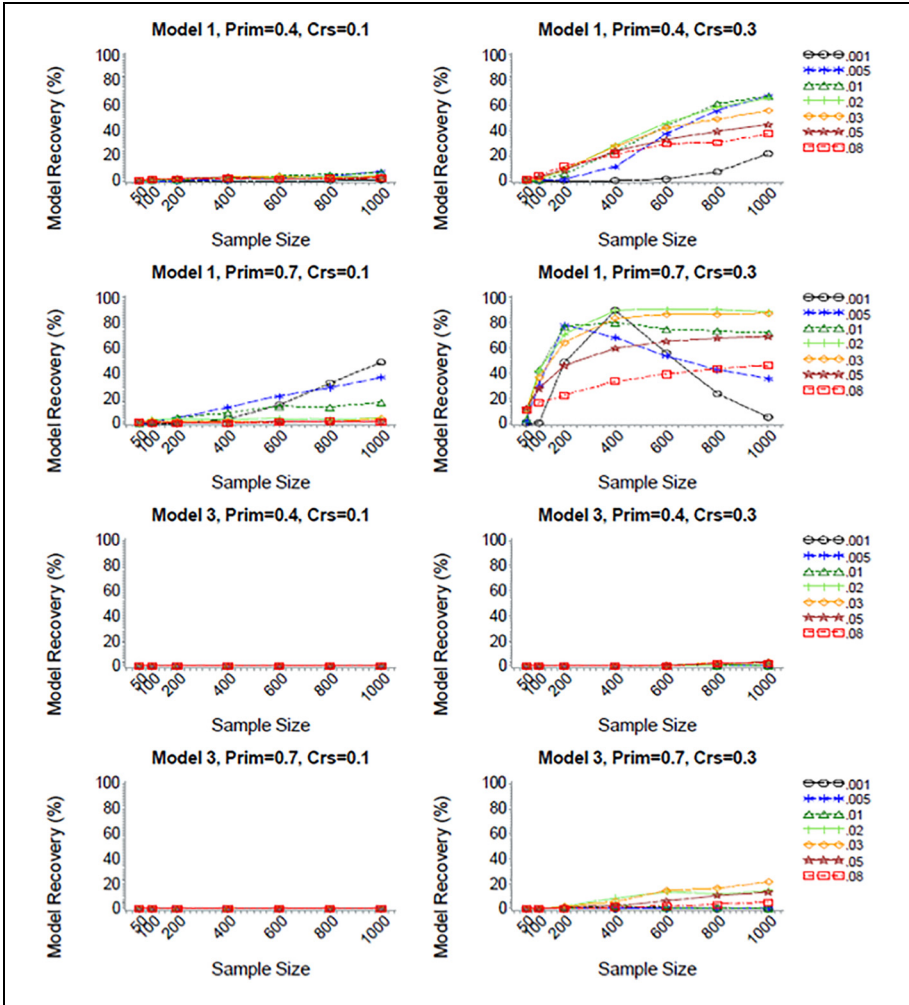


Figure 1. Model recovery rates.

Note. The legend shows the prior variance; Model 1 has one cross-loading; Model 3 has four cross-loadings; Prim = primary loadings; Crs = cross-loadings.

obvious increase (due to the increase in power) and then a decrease (due to the increase in false positives). The decrease in recovery rates indicated that extra cross-loadings may be detected in addition to the population cross-loadings, particularly, with large sample sizes. This is in line with results for true positive rates (Figure 2) and false positive rates (Figure 3). In relatively large samples with small prior variances, while the true positive rates constantly increased, the false positive rates were also elevated.

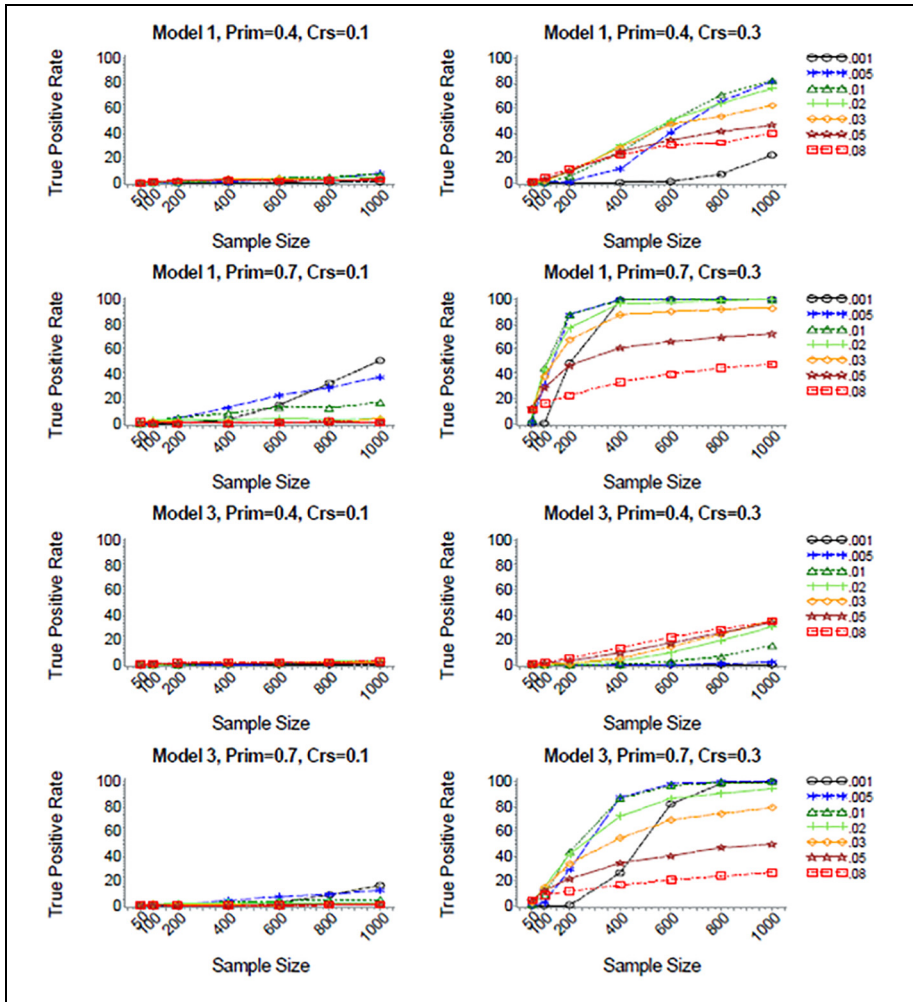


Figure 2. True positive rates: concluding nonzero cross-loadings as nonzero.
Note. The legend shows the prior variance; Model 1 has one cross-loading; Model 3 has four cross-loadings; Prim = primary loadings; Crs = cross-loadings.

Parameter Estimates and 95% Credible Intervals. Figures 4 to 6 show the parameter estimates and 95% CIs for primary loadings, cross-loadings, and the factor correlation, respectively. In each condition, the mean parameter estimates across replications for the seven sample sizes (50-1,000) were shown by seven prior variances (.001-.08). Each mean parameter estimate was plotted with a bar indicating the 95% CI.

Overall, the 95% CIs of parameter estimates shrunk as the sample size increased and/or prior variance decreased. The estimation for primary loadings were relatively

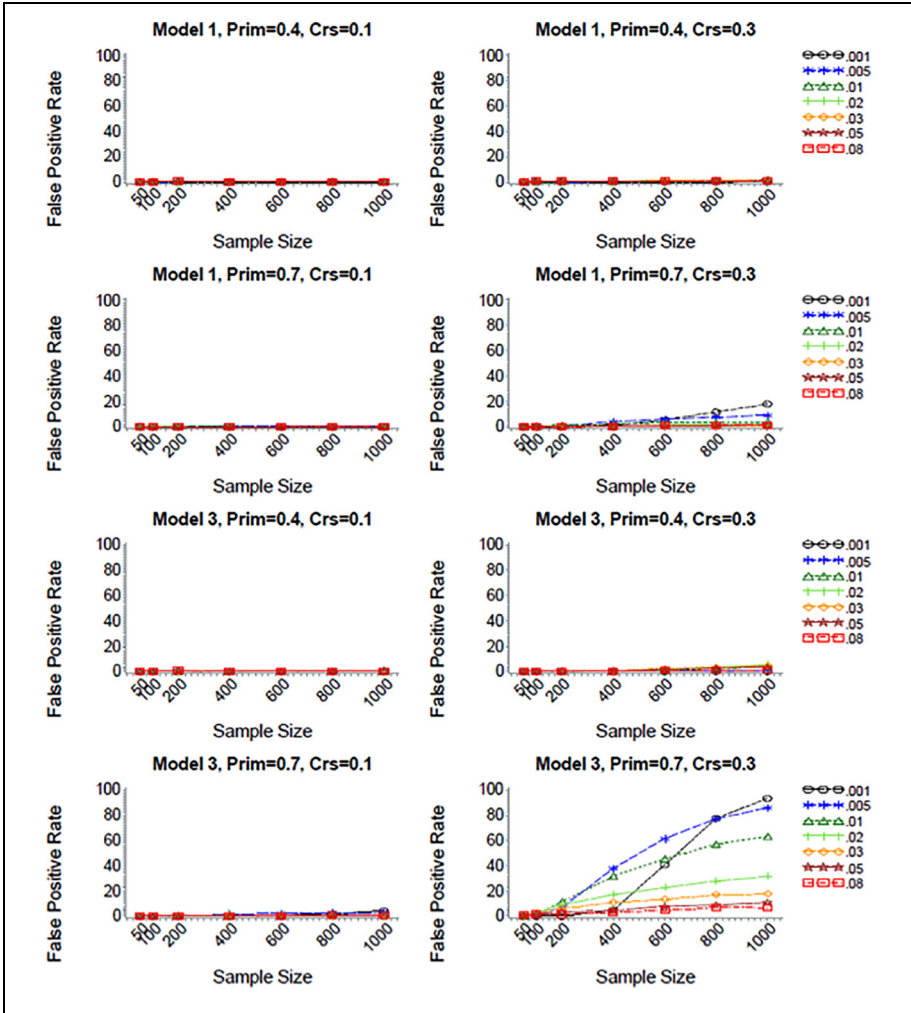


Figure 3. False positive rates: concluding zero cross-loadings as nonzero.

Note. The legend shows the prior variance; Model 1 has one cross-loading; Model 3 has four cross-loadings; Prim = primary loadings; Crs = cross-loadings.

the most accurate, compared with the estimation of cross-loadings and the factor correlation. Only when the sample size was 50 did the estimates slightly deviate from the true primary loading value. For cross-loadings, an underestimation was observed when the true cross-loading was equal to .3. In these conditions, the underestimation was more severe with smaller prior variances. This phenomenon was partly due to the zero-mean priors imposed on the cross-loadings, which pulled the center of posterior distributions toward zero. It was also partly due to the light tails of the normal

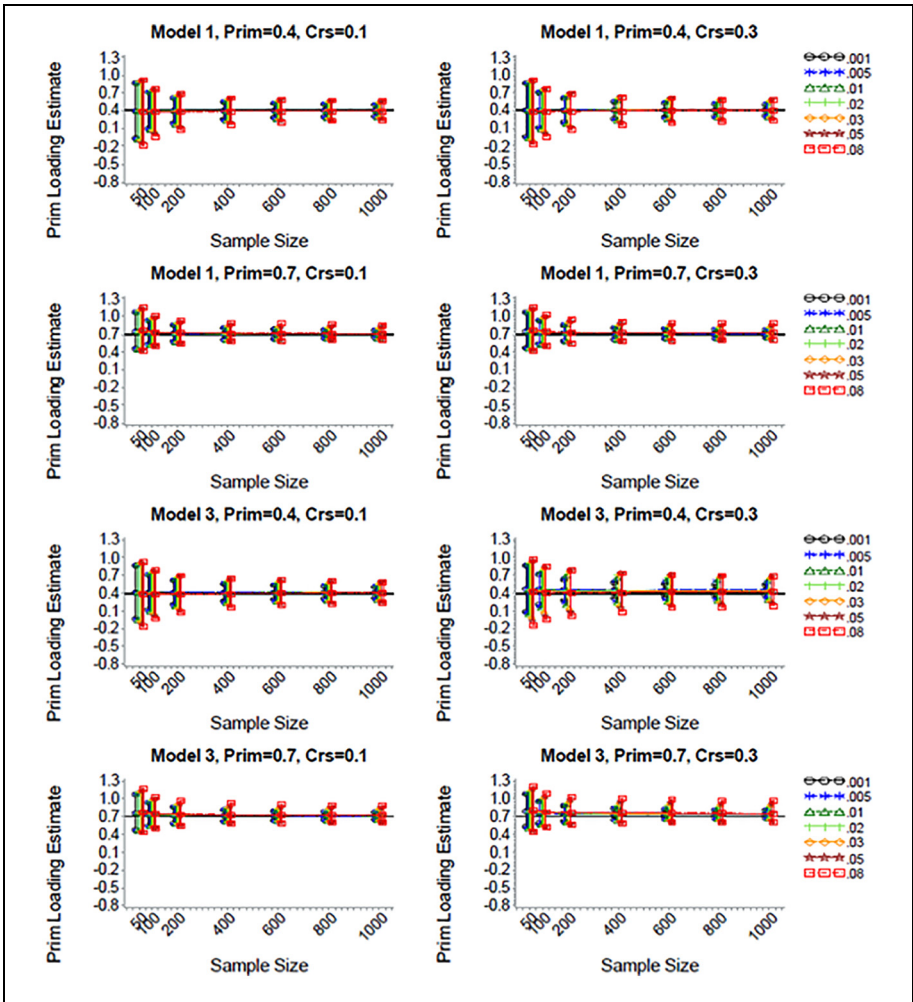


Figure 4. Parameter estimates and 95% credible intervals for primary loadings.
Note. The legend shows the prior variance; Model 1 has one cross-loading; Model 3 has four cross-loadings; Prim = primary loadings; Crs = cross-loadings; the vertical bars indicate the 95% credible intervals.

distributions with smaller prior variances; a heavier-tail prior may alleviate the underestimation but could potentially induce convergence problems in complex models. As the sample size increased, the underestimation was alleviated because the data likelihood based on a larger sample mitigated the shrinkage of parameters toward zero. As cross-loadings were forced to shrink toward zero, factor correlation estimates were also affected as being overestimated in most conditions, especially with the true cross-loading of .3. When the number of cross-loadings increased, the

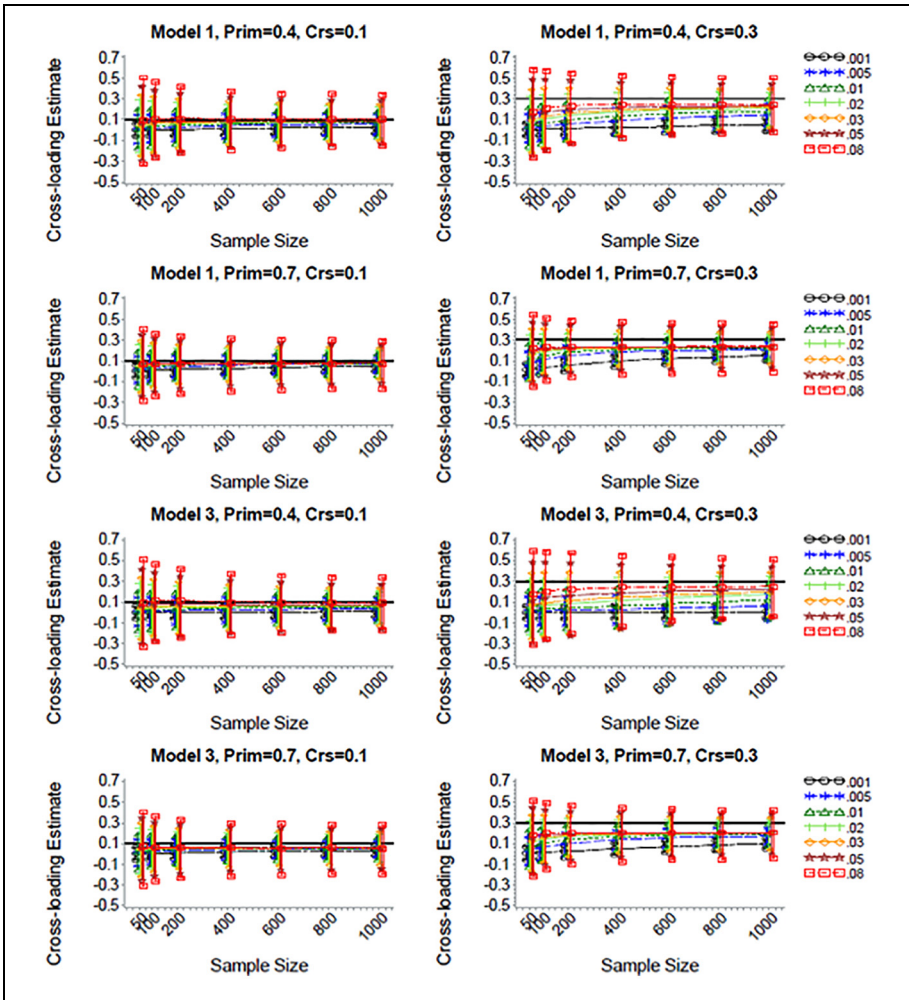


Figure 5. Parameter estimates and 95% credible intervals for cross-loadings.

Note. The legend shows the prior variance; Model 1 has one cross-loading; Model 3 has four cross-loadings; Prim = primary loadings; Crs = cross-loadings; the vertical bars indicate the 95% credible intervals.

overestimation was more severe. For example, in Model 3 with primary loading of .4, the factor correlation estimates were severely inflated up to around .8 (the true factor correlation was .4).

In addition, the coverage rates of 95% CIs decreased as the prior variance decreased in general. The coverage rates were low for cross-loadings and the factor correlation when the cross-loading was .3 and the prior variance was smaller than

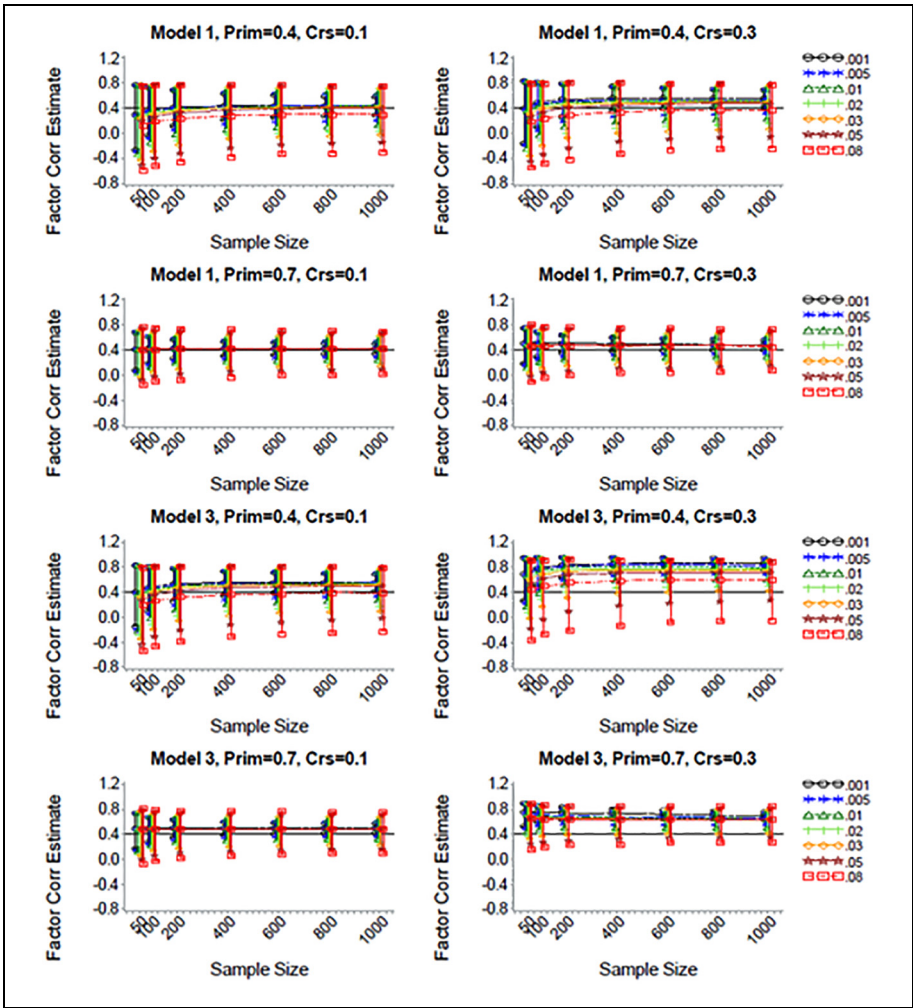


Figure 6. Parameter estimates and 95% credible intervals for the factor correlation.
Note. The legend shows the prior variance; Model 1 has one cross-loading; Model 3 has four cross-loadings; Prim = primary loadings; Crs = cross-loading; Factor Corr = Factor Correlation; the vertical bars indicate the 95% credible intervals.

.005. The coverage rates for primary loadings, cross-loadings, and the factor correlation can be found in the online Supplemental Materials.

Results for Noninformative/Vague Prior Sensitivity

The sensitivity to noninformative/vague priors was evaluated for one condition—Model 1, sample size of 50, primary loading of .7, and one cross-loading of .3. Table 3 shows

Table 3. Sensitivity of Model Fit and Variable Selection to Noninformative/Vague Priors—Model 1, Primary Loadings of .7, One Cross-Loading of .3, and Sample Size of 50.

Info prior variance on cross-loadings	Priors on other parameters	PPp rejection rate (%)	PPPp rejection rate (%)	Model recovery rate (%)	True positive rate (%)	False positive rate (%)
.001	1. <i>Mplus</i> default	6.6	6.8	.0	.0	.0
	2. <i>IG</i> (-.5, 0)	4.8	6.2	.0	.0	.0
	3. <i>IG</i> (.01, .01)	4.0	5.0	.0	.0	.0
	4. <i>IG</i> (.1, .1)	4.4	5.6	.0	.0	.0
	5. Vague normal	6.2	5.8	.0	.0	.0
.01	1. <i>Mplus</i> default	1.2	2.2	3.4	3.4	.0
	2. <i>IG</i> (-.5, 0)	1.0	2.2	5.4	5.4	.0
	3. <i>IG</i> (.01, .01)	.8	2.2	6.0	6.0	.0
	4. <i>IG</i> (.1, .1)	.8	1.6	6.4	6.4	.0
	5. Vague normal	1.6	2.8	3.4	3.4	.0
.03	1. <i>Mplus</i> default	.8	.2	10.8	11.6	.2
	2. <i>IG</i> (-.5, 0)	.4	.2	12.8	13.8	.2
	3. <i>IG</i> (.01, .01)	.4	.0	14.4	15.0	.3
	4. <i>IG</i> (.1, .1)	.0	.0	13.6	14.2	.3
	5. Vague normal	.8	.2	10.8	11.6	.2
.08	1. <i>Mplus</i> default	.6	.0	11.0	11.4	.2
	2. <i>IG</i> (-.5, 0)	.6	.0	10.8	11.4	.3
	3. <i>IG</i> (.01, .01)	.6	.0	11.8	12.2	.3
	4. <i>IG</i> (.1, .1)	.8	.0	10.6	11.2	.3
	5. Vague normal	.6	.0	11.0	11.4	.2

Note. Info prior variance = informative prior variance; PPp = posterior predictive *p* value; PPPp = prior-posterior predictive *p* value. Item intercepts and factor loadings have a normal distribution $N(0, \infty)$, except for the vague normal prior. *Mplus* default combines $N(0, \infty)$ for intercepts and loadings and $IG(-1, 0)$ for residual variances. *IG*(-.5, 0) = noninformative improper prior; *IG*(.01, .01) and *IG*(.1, .1) = vague proper priors. Vague normal prior combines $N(0, 1000)$ for intercepts, $N(0, 100)$ for loadings, and $IG(-1, 0)$ for residual variances.

the sensitivity of model fit and variable selection results to noninformative/vague priors. Because result patterns were similar across the seven shrinkage priors, only results using prior variances of .001, .01, .03, and .08 on cross-loadings were reported. Based on both PPp and PPPp rejection rates, the use of *Mplus* default priors (Prior 1) and vague normal priors (Prior 5) appeared to produce slightly higher rejection rates than the use of other priors in most conditions. Though not reported in the table, model fit indices and ICs were not affected much by different noninformative/vague priors. The use of vague proper priors (*IG*(.01, .01) and *IG*(.1, .1)) yielded slightly higher model recovery rates and true positive rates, when the shrinkage prior variances were between .01 and .05. The false positive rates were comparable across all choices of noninformative/vague priors.

Table 4 shows the sensitivity of parameter estimation to noninformative/vague priors. In all conditions, the use of various noninformative/vague priors had little impact on model parameter estimates and lower and upper bounds of the 95% CIs.

Table 4. Sensitivity of Posterior Estimates and 95% Credible Intervals to Noninformative/Vague Priors—Model 1, Primary Loadings of .7, One Cross-Loading of .3, and Sample Size of 50.

Info prior variance on cross-loadings	Priors on other parameters	Primary loading (pop = .7)			Cross-loading (pop = .1)			Factor correlation (pop = .4)		
		Estimate	Lower 2.5%	Upper 2.5%	Estimate	Lower 2.5%	Upper 2.5%	Estimate	Lower 2.5%	Upper 2.5%
.001	1. Mplus default	.75	.47	1.07	.01	-.05	.08	.51	.19	.75
	2. $IG(-.5, 0)$	.75	.47	1.07	.01	-.05	.08	.51	.19	.75
	3. $IG(.01, .01)$	.75	.48	1.07	.02	-.05	.08	.51	.19	.75
	4. $IG(.1, .1)$	.76	.48	1.07	.02	-.05	.08	.51	.18	.74
	5. Vague normal	.75	.47	1.07	.01	-.05	.08	.51	.19	.75
.01	1. Mplus default	.75	.46	1.08	.10	-.08	.27	.49	.13	.75
	2. $IG(-.5, 0)$	.75	.46	1.08	.10	-.07	.27	.49	.13	.75
	3. $IG(.01, .01)$	.76	.47	1.08	.10	-.07	.27	.48	.13	.74
	4. $IG(.1, .1)$	.76	.47	1.08	.10	-.07	.27	.48	.13	.74
	5. Vague normal	.75	.46	1.08	.10	-.08	.27	.49	.13	.75
.03	1. Mplus default	.76	.45	1.11	.17	-.09	.41	.47	.04	.77
	2. $IG(-.5, 0)$	.76	.45	1.11	.17	-.09	.41	.47	.04	.76
	3. $IG(.01, .01)$	.77	.46	1.11	.17	-.09	.41	.46	.04	.76
	4. $IG(.1, .1)$	.77	.46	1.11	.18	-.08	.41	.46	.04	.76
	5. Vague normal	.76	.45	1.11	.17	-.09	.41	.47	.04	.77
.08	1. Mplus default	.78	.43	1.16	.22	-.14	.55	.46	-.10	.81
	2. $IG(-.5, 0)$	.78	.44	1.16	.22	-.14	.55	.46	-.09	.80
	3. $IG(.01, .01)$	.78	.44	1.16	.22	-.14	.55	.46	-.09	.80
	4. $IG(.1, .1)$	.78	.45	1.16	.23	-.13	.55	.45	-.10	.80
	5. Vague normal	.78	.43	1.16	.22	-.14	.55	.46	-.10	.81

Note. Info prior variance = informative prior variance; pop = population value; Lower 2.5% = 2.5 percentile of the posterior distribution; Upper 2.5% = 97.5 percentile of the posterior distribution; Item intercepts and factor loadings have a normal distribution $N(0, \infty)$, except for the vague normal prior. Mplus default combines $N(0, \infty)$ for intercepts and loadings and $IG(-1, 0)$ for residual variances. $IG(-.5, 0)$ = noninformative improper prior; $IG(.01, .01)$ and $IG(.1, .1)$ = vague proper priors. Vague normal prior combines $N(0, 1000)$ for intercepts, $N(0, 100)$ for loadings, and $IG(-1, 0)$ for residual variances.

Study 2: An Empirical Example

Method

The prior sensitivity may be better understood through an analysis of real data. A subset of data was used from the client survey of Arkansas Rehabilitation Services' (ARS) Comprehensive Needs Assessment (Williams & Vire, 2019). The purpose of the assessment is to determine whether ARS is providing the right services, the services that are needed by the citizens who have disabilities, and whether the quality of those services is sufficient to actually meet the needs of the clients. The survey was administered to people who are or have been the clients of ARS. A total of 623 participants were included in the analysis, after excluding the 38 cases with missing data on all variables.

In this survey, fifteen items were intended to measure three correlated latent constructs: (1) readiness and placement (ξ_1) measured by items y_1 - y_3 , (2) barriers to service (ξ_2) measured by items y_4 - y_{11} , and (3) service efficacy (ξ_3) measured by items y_{12} - y_{15} . The actual survey items are available in Appendix A. Each item was measured on a 5-point Likert-type scale, in which the response categories were *strongly agree*, *agree*, *not decided*, *disagree*, and *strongly disagree*. An ICM-CFA model analyzed using BSEM with all noninformative priors has shown acceptable model fit ($PPp < .001$, BCFI = .934, BTLI = .920, BRMSEA = .074).⁴ In an attempt to further improve model fit and identify potential cross-loadings, BSEM-N analyses with various shrinkage priors on cross-loadings were conducted on the basis of the three-factor model. Asparouhov et al. (2015) suggested to first specify a small enough prior variance on cross-loadings, and then gradually increase the prior variance and observe changes in model fit and parameter estimation. The shrinkage priors used on cross-loadings were the same as those used in the simulation study. That is, the shrinkage priors had a mean of zero, and a variance of .001, .005, .01, .02, .03, .05, or .08, respectively. In addition, an exploratory factor analysis (Jennrich & Sampson, 1966) with the oblique Geomin rotation (Yates, 1987) was also conducted in comparison to results from BSEM-N with shrinkage priors.

Results

The reporting was focused on model fit and significant standardized cross-loadings, as shown in Table 5. Comparing between PPp and $PPPp$, PPp was not sensitive to the prior variances and suggested rejection of the model regardless of the priors used. On the other hand, the $PPPp$ value increased as the prior variance increased, and suggested rejection of the model only when the prior variances were .005 and lower. Most other fit measures were not sensitive to the prior variances. The changes in fit measures were small: $\Delta BCFI \leq .003$, $\Delta BTLI \leq .009$, $\Delta BRMSEA \leq .008$ and $\Delta DIC \leq 4$, except for $\Delta BIC \leq 24$. Comparing with BSEM-N models, the exploratory factor analysis provided comparable values of frequentist CFI, TLI, and RMSEA, but notably lower BIC value. Given the above evidence, the BSEM-N

Table 5. Model Fit and Significant Cross-Loadings in the Empirical Example.

Model fit						Significant cross-loadings						
						ξ_1 by			ξ_3 by			
PPp	PPPp	CFI	TLI	RMSEA	DIC	BIC	γ_9	γ_{15}	γ_3	γ_4	γ_8	γ_{11}
BSEM-N prior variance on cross-loadings												
.001	<.001	.948	.929	.070	21,279	21665	—	—	—	.046	—	.065
.005	<.001	.951	.924	.073	21,276	21644	—	.071	—	.085	—	.111
.01	<.001	.951	.922	.074	21,277	21642	—	.082	—	.102	—	.129
.02	<.001	.950	.920	.075	21,279	21641	—	—	—	.107	—	.129
.03	<.001	.950	.920	.075	21,280	21642	—	—	—	—	—	.130
.05	<.001	.950	.920	.074	21,280	21642	—	—	—	—	—	—
.08	<.001	.950	.922	.074	21,279	21643	—	—	—	—	—	—
EFA	—	.947	.912	.078	—	21601	.101	.089	-.071	.127	.074	.152

Note. PPPp = posterior predictive p value; PPPp = prior–posterior predictive p value; RMSEA = root mean square error of approximation; CFI = comparative fit index; TLI = Tucker–Lewis index; BIC = Bayesian information criterion; DIC = deviance information criterion. For BSEM, the CFI, TLI and RMSEA are Bayesian variants of fit indices. ξ = latent factor; γ = observed item. EFA is exploratory factor analysis with the Geomin rotation.

models with $PPPp > .05$ were taken into consideration, which were models with prior variances of .01 to .08.

Examining the primary loading values across the candidate models (with prior variances of .01 to .08), the standardized primary loadings ranged from .63 to .95, with most estimated between .70 and .90, indicating strong associations between the items and latent factors. The significant cross-loadings were estimated between .046 and .130, indicating relatively small cross-loadings. Based on the simulation results, when the number of true cross-loadings was small, we expect relatively accurate estimates of all model parameters. In the current example, if the purpose is to obtain good parameter estimates, the model with a prior variance of .02 or .03 may be considered, because it suggested a small number of significant cross-loadings, allowed more flexibility in cross-loading estimates than models with prior variances less than .02, while providing smaller 95% CI of parameter estimates comparing with models with prior variances greater than .03. With prior variances of .02 or .03, the prior 95% CIs were $\pm .28$ and $\pm .34$, respectively. Both covered the largest estimated cross-loadings well.

If the purpose is to identify important cross-loadings in the model, suggestions from models with prior variances of .01 to .03 may be considered. Based on the simulation results, when cross-loadings were small, we expect that false positives are low and model recovery rates increase as sample sizes increase. Given the small cross-loading estimates in this study, it indicates that the identified significant cross-loadings are less likely to be false positives. As we have multiple cross-loadings suggested by both BSEM-N models and the EFA model, we may consider respecifying the ICM-CFA model by relaxing one significant cross-loading at a time. Then a model comparison can be conducted between the simpler and more complex model (with an added cross-loading) to examine whether the more complex model fits significantly better than the simpler model. This process can be repeated until the two completing models are not statistically different. More specifically, in the given scenario, when deciding which cross-loadings to be added to the model, all models (with prior variances from .01 to .03) suggested adding the cross-loading of y_{11} on ξ_3 . Therefore, the cross-loading of y_{11} on ξ_3 can be first added to the revised model. If the revised model fitted better than the ICM-CFA, the next cross-loadings to be added sequentially would be y_4 on ξ_3 and y_{15} on ξ_1 . This method, however, is limited in that the same data were used for both the variable selection and model modifications.

Discussion

This study evaluated the prior sensitivity using BSEM-N as a variable selection and model estimation tool in factor analysis with sparse factor loading structures via a simulation study and an empirical example. The sensitivity to both the shrinkage priors and noninformative/vague priors was examined based on model fit, model recovery rates, true and false positive rates, and parameter estimation. Major findings are summarized and discussed as follows.

For the model fit evaluation, the PPP_p was more sensitive to the shrinkage prior variances than the PP_p in BSEM-N. The PPP_p tended to reject less often than the PP_p when the prior variance was large and more often when the prior variance was small. In line with Hoijtink and van de Schoot (2018), the PPP_p was more powerful than the PP_p not only in regression but also in factor analysis using BSEM-N with small-variance priors. When both the primary and cross-loadings were small, DIC may be inconsistent with other fit measures. BCFI, BTLI, and BRMSEA were insensitive to choices of both shrinkage priors and noninformative/vague priors.

The shrinkage prior variance on cross-loadings clearly played a role on variable selection. It appeared that the prior worked well when its 95% credible interval barely covered the population cross-loadings. More explicitly, based on the simulation findings, with a cross-loading of .1, the prior variance of .005 (95% CI: $\pm .14$) and .001 (95% CI: $\pm .06$) resulted in the best balance between model recovery, true positive and false positive rates. With a cross-loading of .3, the prior variance of .02 (95% CI: $\pm .28$) and .03 (95% CI: $\pm .34$) outperformed other prior choices; the commonly used prior variance of .01 led to good true positive rates but at the cost of inflating false positives. Nonetheless, the use of shrinkage priors comes with a bias-variance trade-off (e.g., Jacobucci et al., 2019). The prior variance appropriate for variable selection may not necessarily yield the best parameter estimates. For parameter estimation, prior variances had the most impact on cross-loading and factor correlation estimates, and the least impact on primary loading estimates. The use of larger prior variances imposed less parameter constraints and provided more accurate parameter estimates than the use of smaller prior variances. Therefore, to improve parameter estimates, a relatively large prior variance may be preferred. With that said, one should also be aware of the consequences of using large prior variances such as the potential model nonconvergence and the difficulty in model interpretations.

The noninformative/vague priors showed a very small impact on model recovery rates, and nearly no impact on parameter estimates and 95% CIs with a sample size of 50. It appeared that the vague proper priors (Priors 3 and 4) resulted in slightly higher model recovery rates (up to 3.6% higher than other priors). The differences among various noninformative/vague priors may diminish as the sample size increases, as indicated in van Erp et al. (2018). Nonetheless, as the noninformative/vague priors were only examined for one condition in this study, their sensitivity in variable selection in other conditions would still be of research interest.

The magnitude of primary loadings also affected various outcomes evaluated in this study. Higher model recovery rates and true positive rates were associated with greater primary loading values. Inflated false positive rates were observed for the primary loading of .7. With larger primary loading values, parameter estimates were more accurate and the 95% CIs were narrower. Hence in practice, if the measurement scales are less reliable (relatively lower associations between indicators and factors), BSEM-N may be less appropriate for variable selection, but could still be considered to improve model estimation, especially with relatively large sample sizes. For variable selection with low primary loading values, other more sophisticated shrinkage

methods (see, Jacobucci & Grimm, 2018) may be considered such as the Bayesian lasso (Park & Casella, 2008) and Bayesian adaptive lasso (Feng et al., 2017).

For empirical researchers, the choice of priors depends on the research purposes. If the goal is to perform variable selection using BSEM-N, a sparse cross-loading structure is required, preferably with a minimal number of nontrivial cross-loadings (e.g., one cross-loading of .3) and relatively high primary loadings. The selection of shrinkage priors largely depends on the magnitude of the cross-loadings. In practice, the cross-loading estimates could be roughly obtained and compared through a sensitivity test for a number of prior choices. Priors with the credible interval boundary values similar to the size of the cross-loading may be given primary consideration. BSEM-N is not recommended for variable selection when many cross-loadings are present (inflated false positive rates with large sample sizes) and/or if the sample size is small (low model recovery and true positive rates).

For the purpose of improving model estimation, BSEM-N works well when the cross-loadings are small (e.g., .1). The number of nonzero cross-loadings and the prior variances have little impact on the parameter estimation. BSEM-N with zero-mean priors is not recommended when the cross-loadings are relatively large (e.g., .3), especially for the estimation of cross-loadings and factor correlations. In this case, the two-step procedure recommended by Muthén and Asparouhov (2012) may be taken into consideration. That is, BSEM-N can first be used to determine the factor loading pattern, and then the revised model is refitted using BSEM-N with noninformative or accurate informative priors to obtain final Bayesian inferences. Nevertheless, as pointed out by Lu et al. (2016), the two-step approach is limited in that the same data are used in both steps which may lead to underestimation of the standard errors. Cross-validation as a remedial approach may be worth investigating further.

Limitations and Future Research

This study is limited by the simulation conditions manipulated. Specifically, only two-factor models were examined. Increasing the model complexity would offer useful insights as it is common that psychological instruments have more than two latent constructs. Other design factors that are worth investigations include the variation in cross-loading patterns, correlated error structures, and factor correlation values. BSEM has many other applications such as estimating models with correlated errors (Guo et al., 2019; Muthén & Asparouhov, 2012) and studying approximate measurement invariance (Liang et al., 2018; Muthén & Asparouhov, 2013; Shi et al., 2017). Examining the prior sensitivity in these areas would be a useful direction for future research. Further, although research has shown that the distributions of ordinal data have little impact on parameter estimates using BSEM-N (Liang et al., 2020), it is not clear whether the distributions of data would have an impact on variable selection. The sensitivity of the distributions (e.g., normal, skewed) and types of data

(e.g., continuous, ordinal) to various shrinkage priors is unknown and warrants further investigation.

It would be useful if the model selection among a set of prior variances can be guided by the comparison of model fit measures. In addition to the fit measures examined in the current study, the fully Bayesian model comparison criteria have been proposed such as the widely available information criterion (WAIC; Watanabe, 2010) and leave-one-out cross-validation (LOO; Vehtari et al., 2017). The use of these Bayesian fit measures to guide model comparisons has emerged in the field of SEM (Liang & Luo, 2019; Lu et al., 2017) and is worthy of investigation.

Researchers should be aware that the choice of priors depends on a number of practical factors, such as the scale reliability, prior belief of parameters, and model complexity. When applying BSEM, a sensitivity test is always recommended to assess the prior impact on the modeling outcomes (see directions for sensitivity tests in Depaoli & van de Schoot, 2017). Nonetheless, recent advances in shrinkage priors have proposed the use of hierarchical models to reduce the number of priors tested in a sensitivity test. In the hierarchical approach, the hyperparameters in a prior are assigned its own prior distributions, namely, hyper priors. Assigning hyper priors allows for the test of a range of hyperparameter values and improves the efficacy of Bayesian estimation. Popular hierarchical shrinkage priors for parameter selection and estimation include adaptive lasso and horseshoe priors (Carvalho et al., 2010). These modern priors have been applied in the regression context, and an extension to the factor analysis models would be of research interest. On the other hand, as the frequentist counterpart to the shrinkage priors, regularized SEM in a frequentist framework has also been proposed to perform variable selection and estimation (Huang, 2018; Huang et al., 2017; Jacobucci et al., 2016; Liang & Jacobucci, 2019). The performance of various regularization methods in factor analysis could be another line of research.

Appendix A

Table A1. Fifteen Questions From the Client Survey of Arkansas Rehabilitation Services' Comprehensive Needs Assessment.

<i>Readiness and placement</i>	
y1	You have the education or training to achieve your employment goals.
y2	You have the job skills to achieve your employment goals.
y3	You have the job search skills to achieve your employment goals.
<i>Barriers to service</i>	
y4	Limited accessibility to ARS via public transportation has made it difficult for you to access ARS services.
y5	Other challenges related to the physical location of the ARS office have made it difficult for you to access ARS services.
y6	ARS's hours of operation have made it difficult for you to access ARS services.
y7	A lack of information about the ARS services available has made it difficult for you to access ARS services.

(continued)

Table A1. Continued

y8	A lack of disability-related accommodations has made it difficult for you to access ARS services.
y9	Difficulties scheduling meetings with your counselor have made it difficult for you to access ARS services.
y10	Other difficulties working with ARS staff have made it difficult for you to access ARS services.
y11	Difficulties completing the Individualized Plan for Employment have made it difficult for you to access ARS services.
<i>Service efficacy</i>	
y12	Your counselor referred you to the Arkansas Career Training Institute (ACTI) for services.
y13	You were accepted for services at ACTI.
y14	You chose to attend ACTI.
y15	The training program that you received at ACTI was valuable.

Note. ARS = Arkansas Rehabilitation Service.

Appendix B

Sample Mplus Code for Implementing BSEM-N

title: BSEM-N for two-factor model

montecarlo:

names = x1-x10;

nobs = 50;

nreps = 500;

results = res.dat; ! save model estimation results

bparameters = bpost.dat; ! save posterior samples

model population:

f1 by x1-x5*.7 x6-x10@0;

f2 by x1-x5@0 x6-x10*.7;

f2 by x5*.3; ! population cross-loading

f1-f2@1;

f1 with f2*.4;

x1-x4*0.51;

x5*0.252;

x6-x10*0.51;

analysis:

estimator = Bayes;

process = 2;

biterations = 100000(1000);

model:

f1 by x1-x5*.7 (p1-p5);

f2 by x6-x10*.7 (p6-p10);

```
[x1-x10] (i1-i10);
f1-f2@1;
f1 with f2*.4;
x1-x4*0.51 (r1-r4);
x5*0.252 (r5);
x6-x10*0.51 (r6-r10);
f1 by x6-x10*0 (c1-c5);
f2 by x1-x4*0 x5*.3 (c6-c10);
```

model priors:

```
i1-i10~N(0,1000); ! priors on intercepts
p1-p10~N(0,100); ! priors on primary loadings
c1-c10~N(0,0.001);! priors on cross-loadings
r1-r10~IG(-1,.0); ! priors on residual variances
```

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ORCID iD

Xinya Liang  <https://orcid.org/0000-0002-2453-2162>

Supplemental Material

Supplemental material for this article is available online.

Notes

1. BSEM can also be implemented using other popular computer programs for Bayesian estimation, including the R package *blavaan* (Merkle & Rosseel, 2018), JAGS (Plummer, 2003), and Stan (Stan Development Team, 2018).
2. A model can be theoretically or empirically underidentified. See more detail about model identification in Kenny and Milan (2012) and Green and Yang (2018).

3. Using *Mplus* default settings, the convergence based on the PSR procedure may be fulfilled in as few as a couple of hundreds of iterations (depending on conditions). However, the PSR factor could still increase after reaching the convergence criterion (Muthén, 2010). To ensure that true convergence was achieved, the minimum iteration for each chain was set to 1,000.
4. *Mplus* does not report PPPp for conventional BSEM. The PPPp is useful and reported when small-variance priors are used.

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